



BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

Database is organised, gathered, structured data that is electronically stored in a computer system.



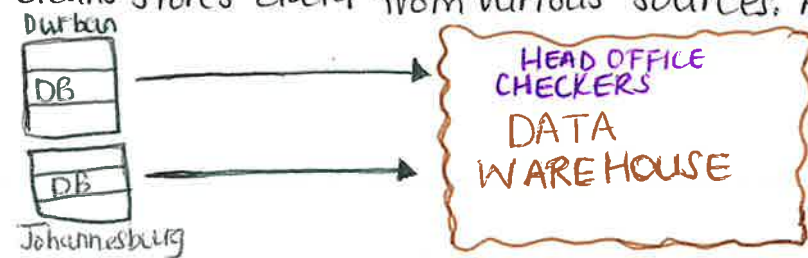
In modern world database is crucial for analysing business trends and patterns, storing vast amounts of data, help with decision making.

Health care: Where all hospitals retain information on test results, Patient visits

Retail Shops: maintain track record of employee information, daily transactions inventory to verify quantity of goods sold or unsold.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse? Data warehouse is a central system for organising, cleaning and storing historical data. A data warehouse maintains a broader system that cleans stores data from various sources. A data base stores day to day operations

checkers Durban & Johannesburg each have a database where every transaction is stored. At midnight the individual Durban & Johannesburg data base is captured in a data warehouse.



3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

Data lake is a reservoir for all types of data, including unstructured and structured data. All the organisation's raw, unstructured data is contained in a data lake, which stores data indefinitely to use now or future. The data warehouse has structured data that is cleaned and processed.

When analysing and loading massive amount of unstructured data, a corporation would rather rely on a data lake than a data warehouse.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A data mart is a subset of a data warehouse that focuses on particular departments or line of business. Data mart makes specific data available for rapid easy access. A data warehouse is organised and cleansed data. A marketing team would use data mart to analyse customer behaviour.



5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured DATA: Is information organised into neat rows and columns with a fixed format eg. Bank statements, Excel spreadsheet.

Semi-Structured DATA: Information that does not live in a standard relational data base table, but uses metadata or hierarchies to group and separate elements. eg. JSON files, NOSQL documents

Unstructured DATA: Is information that lack a predefined data model or table structure. eg. Videos, social media.



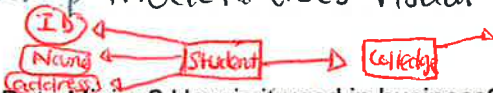
6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models. Data modeling is how organised your data is to make it easy to use, creating a map on what data you have and the connection on how it will be stored. It is vital to prevent duplicate and messy data to make reports and searches faster. The two types of data Modeling are:

Relational
Relational

DATA MODEL & Data is stored in tables and rows, columns

ID	Name
101	Kim
234	Joe

ENTITY-Relationship Model uses Visual diagrams of entity's



7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action. Data Mining is finding hidden patterns in number of data that one would not see by looking at it. Data mining is turning old data into money or better service.

Example: Foschini - sales group will mine your rewards card / purchase history. They see you purchased a Luella bag, foschini sales team will send a SMS or email with a special "get 20% off if you purchase a Luella animal skin wallet."

Streaming: Netflix / Disney+. They mine series or movies you have watched, they offer a "because you have watched option". They give you more series or relevant movies you have not seen as they saw and mined patterns of thousand of people similar to your taste in series.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

Database management systems is the layer of software that stands between the user and the unprocessed data that is stored, which enables speedy data retrieval for user.

The four popular DBMS platforms are:

Data Bricks: Cloud DBMS designed to handle more data. keeps everything, including social media postings, tables, files and videos. Blends structured and unstructured data. warehouse.

Snowflake: This data base system that operates entirely on the cloud. Instead of installing, you log on to your computer, designed to analyse and store large amount of data.

PostgreSQL: Free more advanced and used by universities and tech companies.

SQL Server: Data is stored in table, rows and columns, uses SQL language.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics. Extract, Transform, load (ETL) is a process that takes data from sources, transforms, cleans it before putting data into the data warehouse.

This process is important because analytics require clean data. It serves as a mediator between messy and clean data.

E - Pulls data from source

T - cleans and fixes by removing duplicate, fixes spelling.

L - put clean data into one DBMS like snowflake where you can analyse



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

Data is usually kept in few specialised systems for large-scale processing and quick queries in analytics.

CSV: used for interchange and small/human-facing files. Once data reaches your platform, converts it to a binary format

Parquet: For analytical storage and BI/ML workloads

Avro: When you require strong ~~she~~ schema evolution and streaming.

JSON: A text format that uses nested structures and key-value pairs. Excellent for semi-structured data & APIs but not suitable for large scale analytics.

CSV - use when exchanging data with non-technical users

JSON - would use to ingest API responses or semi-structured.

Avro - will be used when streaming/event pipeline.

Parquet - use storing analytical data.



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

- 11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.**

Artificial Intelligence (AI) is a technology that enables machines and computers to carry out human functions. Narrow AI is poor AI that we encounter on a daily basis trained to complete specific tasks frequently more quickly than human mind. General AI does not exist yet it can do intellectual human tasks.

Narrow AI: Alexa is an AI robot that can connect to gadgets or smart home and be programmed to play music or turn off lights or make phone calls.

General AI: Human-like robot it would have common sense and reasoning.
eg. Humanoid robots.

- 12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.**

Machine learning, instead of depending on explicit programming, machine learning allows systems to learn from data. It finds patterns and enhances experience rather than analysing data.

Traditional programming provides the computer data and rules. It will provide you with the answer, eg. Excel formula. In machine learning, you give the computer data and answer, and it learns the rules on its own. eg. Truecaller detects scams or insurance sales, frauds.

- 13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?**

Deep learning is neural networks brain-like structure that is used in deep learning. It uses only raw data, such as a picture or a video, to determine what to look for on its own, eg. Biometric fingerprint or facial recognition. AI enables robots to behave intelligently like people, however deep learning determines what to look for on its own.

Neural networks are joined to form artificial neurons to neural networks, which mimic the functioning of your brain.

- 14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.**

Natural language processing is a branch of artificial intelligence, processing aids computers in comprehending, interpreting, and producing human language. It acts as a go-between for computer coding and human communication.

Question 12.

The three main types of machine learning:

Supervised learning:

- Gains the ability to anticipate outcomes for new inputs by learning labeled instances (input-output) pair.
eg. classification (spam vs not spam)
forecasting house prices.

Unsupervised learning:

Without predetermined answers, it learn from unlabeled data to uncover hidden structure or patterns.

- eg. when you don't already know the segments, you divide your clients into groups according to their purchasing habits.

Reinforcement learning:

Through interactions with the environment and incentives or penalties for actions, an agent learns to optimise a policy over time.

- eg. Sequential decision-making activities, such as Control Systems, robotics, and gaming.

Question 15

Large language model, You hire a administrator who has all the skills and work experience on processing and approvals of documentation and can ask anything.

Question 16

CHATGPT

- An AI chat bot that uses large models, large language models to understand human text and generate human like responses. used for brainstorming.

GEMINI

- It processes multiple types of media at the same time, including text, audio, images and video. Analyses large files.

MIDJOURNEY

- It turns text prompts into highly detailed, artistic images, using diffusion models. Creates digital art.



The three real world application is:

SIRI - responds to your Spoken Phrase or Command.

Machine translator - Google translate its a tool that converts text from one language to another.

Chatbots - Sars asks "How do I submit" or "how can we help"

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

Large language Model (LLM) a sophisticated artificial intelligence program is made to read, comprehend and produce text that is similar to that of a person. While large language model is a specialised advanced from deep learning intended to comprehend and produce human language, standard machine learning is a broad category of artificial intelligence focused on identifying patterns in structured data to make specific predictions. **Standard machine learning teaches an administrator assistant how to approve and stamp documents. The assistant will be good at approving and stamping documents.**

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

Generative AI is unpredictable new things produced. It can produce speech, photos text and videos. operates by reading everything before you begin to compose a sentence eg. Composing a community announcement on a whatsapp group, it will begin to guess the following words until the message is complete.

Generative AI uses **chatGPT, Gemini, MIDJourney**.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

Training dataset is a collection of data used to teach a machine learning, teaching AI recognised high performing post.

The quality of training data is so important when building an AI model because AI is only as good as what you teach it. If the training data is biased or incorrect then training data will be biased and incorrect, the model learns the wrong patterns and will produce systematically unfair, inaccurate or unsafe outputs. This is the "garbage in, garbage out" problem in machine learning.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer vision is a sub field of artificial intelligence that allows computers to see, through the interpretation and comprehension of images, video and other visual data. Photos must first be converted to numbers before machines can see them. The images are then given meaning such as labels, positions, relationships.



by using layered mathematical models, especially deep neural networks, to extract patterns that range from simple edges to complex objects.

examples:

Media, Sports:

Sport analytics include, performance measures obtained from broadcasting or field cameras, offside/line calls, and player and ball monitoring.

~~One device~~ Vision models are used by photo and video apps for features like auto-tagging, object portrait mode, augmented reality effects.

Automotive Vehicles:

Tesla - Self driving car - to enable lane keeping, automated emergency brakes, on board cameras identify lanes, traffic signs, pedestrians and other vehicles.



Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

Application programming interface is a collection of rules and processes that allow one software program to ask another for information or actions without needing to understand how other programs operate.
Social logins ("Sign in with Google or Apple") use authentication APIs to verify your identity.
API is important for security control, API conceal internal systems, enforce access rules and only reveal authorized actions and data.
Modularity - as long as the API contract remains stable, teams can separately develop and update services.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

A API key is a special, secret code identifies it so the API provider can verify approve and monitor your usage. e.g. Google Maps
Identity for apps: the key tells the API which app or project is making the request not necessary which human use is behind it.
Access control: to determine whether to approve the request and what features or data the caller can access, the API verifies the key.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

Rest API, Representational State Transfer is referred to as REST. An API adheres to the REST architectural style this allows clients and servers to exchange data in a straightforward, consistent manner by utilizing common web protocols, typically ~~the~~ HTTP.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON - JavaScript object notation a lightweight format for structured data such as list and objects that facilitate information interchange between various systems and programming languages. JSON is quick to process and naturally translates to the data structure used in contemporary programming, particularly JavaScript.

Question 20

If the API key is exposed publicly, hackers can use it in a matter of ~~minutes~~ minutes to pretend to be your application, collecting fees, stealing information and misusing your services until you cancel it.

Question 21

A REST API exposes endpoints (URLs) for resources like users, orders, or products. Clients call these endpoints with HTTP methods to perform action.

GET: obtains a resource.

To retrieve a representation of a ~~source~~ resource such as, list, user or product

POST: Build resources and submit data.

The goal is to send data processing, which often results in the creation of new resources or initiation of operations.

PUT: Swap out a resource (whole update)
use given representation to replace the full target resource; if it does not already exist, it may be created, depending on the API design.

DELETE: Get rid of a resource
Removing the designated resource.

Question 24

IaaS: (Infrastructure as a Service)

Rent virtual machine, storage and networks,

E.g. Infrastructure as a Service (IaaS): amazon EC2 (Elastic Compute) is a virtual server that allows you to set up and run your ^{cloud} own operation system and programs.

PaaS: Platform as a Service - rent a ~~man~~ managed platform to build and deploy apps without managing servers. example - Google app engine. Google app engine is a managed platform that allows web programs to be deployed without requiring server or runtime environment management.

SaaS: Software as a Service - use software hosted in the cloud via a browser/app example, Gmail, Microsoft 365, Salesforce.



Objects used - key value pair collections enclosed in brackets `{}` enclosed in double quotations are keys.

eg. `{"name": "age": 30, "active": true}`

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Prompt Engineering a process of meticulously crafting and perfecting the text instructions (prompts) you provide to a generative AI model in order to ensure that it generates precise pertinent and reliable results. With technologies like ChatGPT Prompt engineering is crucial because model merely guesses the next token based on your input. The aim is made clear by well-written prompts. Poor: "Explain homeostasis" Better: "You are a biology"
Better: I am a biology intern in high school for 16 year olds, describe homeostasis in four to five short phrases, finish with 2 questions test with answers.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

Map function is a higher order function that performs a given operation on each element in a collection (lists or array) and returns a new collection of the results without modifying the original data. Each element is transformed when a function is applied. Map applies ~~each~~ the function to each item and gathers the results in the correct order.

Rather than using an explicit loop, it encodes "transform every item" in single line function style - makes data transformations composable and simple to understand by fitting pipelines such as map → filter → reduce

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Cloud Computing is the pay as you go on demand supply of computer resource via the Internet, including servers, storage, database, networking and application. Example: Microsoft 365 is a cloud-hosted suite of productivity applications that can be accessed through desktop clients or browsers, including Word, Excel and Outlook.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Version Control System is a software program that keeps track of changes made to files, particularly source codes. This allows you to collaborate without overwriting each others work, view history and fix errors. e.g. 'Save as' ~~word~~ on a word document. GIT is a distributed version control system that allows you to save snapshots, undo errors and work together without overwriting each others work by tracking changes to files



commit - a photo of your code / Project at a moment
Branch - a separate timeline where you keep taking new photos without affecting the main timeline, later you can merge, timeline back.

A Commit is a snapshot of your project that has been saved at a specific moment in time. It becomes a permanent point in the project's history and documents what changed. Who made the change, when and why via a Commit message.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

A Jupyter notebook is an open source, browser-based tool, allows users to create and share interactive documents that include live code, output/results, visualisation and explanatory text all in one location. Python is far the most popular programming language. Because it consolidates the entire analytical workflow, code, results, charts and explanations, into one single interactive, shareable document. Jupyter is well liked among Analysts and data Scientists.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a high level all purpose programming language renowned for its extensive library ecosystem and straight-forward, legible syntax. Simple, english syntax and indentation blocks make it easy to read and write also quick to learn.

The four Python libraries commonly used in data work :

Multi Paradigm :
 Enables dynamic typing, rich built in data structures, lists, dicts, sets and object oriented, Procedural and functional approaches.

Data-frames : is for cleaning, manipulating, aggregating and merging tabular data are provided by Pandas, a data manipulation and analysis tool.

Matplotlib : is a fundamental charting package that produces static, charts such as Scatter plots, line plots and histograms.

Scikit-learn : is a machine learning toolkit that contains preprocessing, model selection, assessment and algorithm for clustering, regression and classification.



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

The idea behind data privacy, also known as information privacy, is that people should be in charge of how companies gather, use, retain, share, and remove personal information about them. It is important for ethics and trust, legal compliance and risk reduction. The General Data Protection Regulation is a comprehensive data privacy regulation of the European Union that establishes guidelines for collection, use, storage and transfer of personal data access, modification, deletion and objection. The countries that use GDPR in the EU countries like Belgium, France and Norway.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data Ethics is set of moral precepts and rules that control the gathering, storing, processing, sharing and use of data ensuring that procedures are just, open, responsible and considerate of individuals' rights. The three examples of unethical use of data are:
- widespread monitoring and facial recognition misidentification.
- facial recognition technology is used by government & law enforcements to track and identify people in public places without their knowledge.
- Amazon biased hiring algorithm - trained on ten years of predominantly male resumes while job screening job applicants learnt to penalize female candidates and lower resumes.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

Data Governance is a set of rules, responsibilities, procedures and guidelines that specify how a company handles its data across the course of its life cycle ensuring that it is correct, safe, accessible and used appropriately.
The ~~four~~^{three} key components of good ^{data} governance framework are:
People, Policies, Processes, technology.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

Bias AI can provide discriminatory results that undermine confidence and security hurt impacted groups, in hiring, lending, policing, health care. AI systems start when generating results that unjustly favour or penalise particular groups or outcomes that is known as bias. This is caused by skewed data, poor design decisions or human beliefs.



example: When AI model consistently produces unfair predictions or judgement that penalize particular groups based on age, gender or race.

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

ROLE	DEFINITION	What they DO	TOOLS
DATA ANALYST	Turn data into business insights. Take information they analyse, find patterns	Dashboards, Reports ad-hoc analysis	SQL, Excel
DATA SCIENTIST	They Predicts build Models	Predict models, experiments & Prototypes	Python, Stats
DATA ENGINEER	They feed data information build infrastructure make data usable	Data pipelines, warehouse / lakes & ETL	SQL, Cloud data Platforms
AI ENGINEER	Turns AI, into functional applications	AI / LLM - powered apps	Python, LLM Framework

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Business Intelligence is the collection of Procedures and tools that gather, examine and display corporate data in order to transform it into useful insights for improved decision-making. This connects to data sources (spreadsheets, databases, SaaS apps, models that cleans the data and displays it in dashboards, reports and charts.

Power BI and Tableau both assist businesses in making data driven decisions.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

I found Section B AI and machine learning interesting mostly because AI is man made and yet AI is taking over with all the programmes, the robots being manufactured. Some are taking over jobs and AI is advanced, but mostly to learn new AI concepts. The different types of AI and parts of being a data analytic.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

I found API keys and data lake abit difficult to understand

Question 31 - The ~~four~~^{Three} key Components of good data governance framework continue.

People

- People, and rights to make decisions
- The governance Council is responsible for Setting strategy, approving policies Setting Priorities for initiatives and resolving cross domain conflicts

Policies

- Classification and sensitivity, by informed handling guidelines, label data Public, Internal and Confidential.

Technology

- Data catalogue and metadata - Central inventory that includes usage Context, ownership and business vocabulary.

Question 34

Power BI : facilitates live, unified dashboards by connecting to several Sources such as (Excel), (SQL) and (SaaS) and Scheduling refreshes to show leaders the most recent KPIs rather than Outdated Spreadsheets.

Tableau : Helps to visual discovery by using a drag and drop interface to Create interactive charts, maps and dashboards that highlight patterns in Complex data.



but I managed to watch Youtube videos and examples to try grasp everything. Most questions were slightly difficult but through examples and definitions it helped.

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

AI & data could affect my future everyday life and career because personally one relies on AI functions. My personal life, I don't have a diary but I can have my phone voice assistant to give me my schedule of the day, I have meetings during the day, Apple watches that can track my health, career wise robotic AI machines that can do almost everything. From surgery, manufacturing a car and a domesticated house robotic assistant. Which raises concerns of job security. Running out of jobs in the near future.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

Good luck! We believe in you.

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